Algebraic Topology

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1 Introduction

1.1 Recap of IB Topological Spaces

We use algebra to turn topological questions to algebraic questions. This results in a loss of information but lets us prove some results.

Example.

$$\mathbb{R}^n \cong \mathbb{R}^m \implies n = m$$

This is a statement which is very hard to prove otherwise. Likewise we can distinguish topological surfaces with a different number of 'holes' in their surface.

For this section all functions are continuous until mentioned otherwise.

Definition. (Map) A *map* is a continuous function.

Lemma. (Gluing lemma) Let $f : X \to Y$ be a function and $C, K \subset X$ be closed subsets with $C \cup K = X$. Then f is continuous if and only if the restriction of f on C and K is continuous.

Proof. We have that f is continuous if and only if for all closed sets $D \subset Y$ the preimage of D is closed. Since $f^{-1}(D) = (f^{-1}(D) \cap C) \cup (f^{-1}(D) \cap K)$, this holds if and only if $\forall D \subset Y$ closed we have that $f^{-1}(D) \cap C$ and $f^{-1}(D) \cap K$ are closed. Since C and K are closed in X, a subset of C (resp. K) is closed in C (resp. K) if and only if it is closed in X. Hence this holds if and only if $(f|_C)^{-1}(D), (f|_K)^{-1}(D)$ are closed in C and K respectively. Hence this holds if and only if the restriction of f on C and K is continuous. □

Lemma. (Lebesgue number lemma) Let (X, d) be a compact metric space. For any open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ of X there exists a $\delta > 0$ such that for all $x \in X$, $B_\delta(x) \subset U_\alpha$ for some $\alpha \in I$.

Proof. Suppose no such δ existed. Then for all $n > 0$ there is some $x_n \in X$ such that $B_{1/n}(x_n) \not\subset U_\alpha$ for all $\alpha \in I$. The sequence $\{x_n\}$ admits a convergent subsequence due to compactness, so *wlog* replace x_n with its convergent subsequence, and take $x_n \to y$. Then take $\alpha \in I$ such that $y \in U_\alpha$. Then since U_α is open there exists a $r > 0$ such that $B_r(y) \subset U_\alpha$. Take $N > 0$ such that

$$\begin{cases} \frac{1}{N} < \frac{r}{2} \\ d(x_N, y) < \frac{r}{2} \end{cases}$$

which implies that $B_{1/N}(x_N) \subset B_r(y) \subset U_\alpha$ which is a contradiction. □

1.2 Cell complexes

Definition. (Cell attachment) Given a space X and a map $f : S^{n-1} \to X$. The space $X \cup_f D^n := X \sqcup D^n / \sim$ where $x \sim f(x), \forall x \in S^{n-1} \subset D^n$ is the result of attaching an n-cell to X along f.
The image of f is called the *attaching sphere*.

Definition. (Cell complex) A *finite-dimensional cell complex* is obtained by:

(i) Start with a set X^0 with the discrete topology.

(ii) Inductively once X^{n-1} is defined we form X^n by attaching a collection of n-cells along some collection of attachment maps

$$\{f_\alpha : S^{n-1} \to X^{n-1}\}_{\alpha \in I}.$$

i.e. that $X^n = \left(X^{n-1} \sqcup \bigsqcup_{\alpha \in I} D^n_\alpha\right) / \sim$ where we define $x \sim f_\alpha(x)$ for $x \in S^{n-1}_\alpha = \partial D^n_\alpha$.

(iii) $X = X^k$ for some $k \geq 0$. The minimal such k is called the dimension of X.

Remark. In general we can make sense of infinite dimensional cell complexes. Suppose that we have

$$X^0 \subset X^1 \subset \cdots$$

such that

(i) X^0 is discrete,

(ii) X^n is formed by attaching n-cells to X^{n-1}.

Define

$$X = \bigcup_{n \geq 0} X^n$$

with the topology where $U \subset X$ is open $\iff U \cap X^n$ is open in X^n for all n. We say that X is a CW-complex and that X^n is the n-skeleton of X.

2 Homotopy and the fundamental group

2.1 Homotopy

Definition. We denote by I the closed interval $[0, 1]$.

Definition. (Homotopy) Let $f, g : X \to Y$. Then we say that a map $H : X \times I \to Y$ is a *homotopy from f to g* if $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. If such an H exists we say that f is *homotopic* to g and $f \simeq g$.

If $A \subset X$ and $f|_A = g|_A$ then a homotopy H from f to g is relative to A (i.e. a homotopy rel A) if $H(a, t) = f(a) = g(a)$ for all $a \in A$ and $t \in I$. If such an H exists we write $f \simeq_A g$.

Proposition. The relation homotopic rel A defines an equivalence relation.

Proof.

(i) $f \simeq_A f$ trivially by $H(x, t) = f(x)$.

(ii) $f \simeq_A g \implies g \simeq_A f$, by replacing t with $1 - t$.

(iii) Suppose that $f \simeq_A g, g \simeq_A h$. Then let H_1 be the homotopy rel A from f to g and H_2 for g to h. Then define

$$H(x,t) = \begin{cases} H_1(x,2t) & 0 \le t \le \frac{1}{2} \\ H_2(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

Since $H_1(x,1) = g(x) = H_2(x,0)$, H is well-defined, and by the gluing lemma (applied to the closed sets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$) H is continuous. Also $H(a,t) = f(a)$ for all $a \in A$, so H is a homotopy rel A from f to h and therefore $f \simeq_A h$. □

Definition. (Homotopy equivalence) Given a map $f : X \to Y$ we say that f is a *homotopy equivalence* if there exists a $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We call g a *homotopy inverse* of f. If there exists a homotopy equivalence $X \to Y$ then X is *homotopy equivalent* to Y and we write that $X \simeq Y$.

Definition. (Contractible) A space X is *contractible* if X is homotopy equivalent to the one-point space $\{*\}$.

Lemma. Let $f_0, f_1 : X \to Y$ be homotopic maps and $g_0, g_1 : Y \to Z$ also be homotopic maps. Then

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

Proof. We want to show that $g_0 \circ f_0 \simeq g_0 \circ f_1 \simeq g_1 \circ f_1$.

Choose homotopies H from f_0 to f_1 and H' from g_0 to g_1. Then $g_0 \circ H$ is a homotopy from $g_0 f_0$ to $g_0 f_1$. We also have that $H' \circ (f_1 \times \mathrm{id}_I)$ is a homotopy from $g_0 f_1$ to $g_1 f_1$ hence we're done by transitivity. □

Proposition. Homotopy equivalence is an equivalence relation.

Proof.

(i) $X \simeq X$ is obvious from the identity map.

(ii) $X \simeq Y$ then we have $Y \simeq X$ just by switching the roles of f and g.

(iii) $X \simeq Y$ and $Y \simeq Z$. Then suppose f, g are the homotopy equivalence and its inverse for X and Y, and f', g' for Y and Z. Consider the map $f'f$ from X to Z and its candidate homotopy inverse gg'. By the lemma above,

$$gg'f'f \simeq g\mathrm{id}_Y f = gf \simeq \mathrm{id}_X$$
$$f'fgg' \simeq f'\mathrm{id}_Y g' = f'g' \simeq \mathrm{id}_Z$$

Therefore $f'f$ is a homotopy equivalence and $X \simeq Z$. □

Example. Consider the inclusion

$$\tau : S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

with homotopy inverse $r : \mathbb{R}^2 \setminus \{0\} \to S^1$ given by projecting each vector onto its unit vector, $r(x) = x/\|x\|$. Since vector normalisation doesn't do anything to unit vectors one of the directions of composition is obvious: $r \circ \tau = \mathrm{id}_{S^1}$. For the other direction,

$$\mathbb{R}^2 \setminus \{0\} \to S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

we're given the homotopy

$$\begin{aligned} H : (\mathbb{R}^2 \setminus \{0\}) \times I &\to \mathbb{R}^2 \setminus \{0\} \\ (x, t) &\mapsto \frac{x}{t + (1-t)\|x\|} \end{aligned}$$

which is well-defined since the denominator is strictly positive, and is a homotopy from $\tau \circ r$ to $\mathrm{id}_{\mathbb{R}^2 \setminus \{0\}}$. Therefore we have that $S^1 \simeq \mathbb{R}^2 \setminus \{0\}$. Similarly we have that $S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$.

Example. We have that $\mathbb{R}^n$ is contractible. Let $f : \mathbb{R}^n \to \{0\}$ be the constant map and $g : \{0\} \hookrightarrow \mathbb{R}^n$ be the inclusion. Then

$$f \circ g = \mathrm{id}_{\{0\}}$$

so one direction is trivial. For the other direction, $g \circ f : \mathbb{R}^n \to \mathbb{R}^n$ is the constant map $x \mapsto 0$. Define the straight-line homotopy

$$\begin{aligned} H : \mathbb{R}^n \times I &\to \mathbb{R}^n \\ (x, t) &\mapsto tx. \end{aligned}$$

This is continuous since scalar multiplication $\mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ is continuous. We have $H(x, 0) = 0 = (g \circ f)(x)$ and $H(x, 1) = x = \mathrm{id}_{\mathbb{R}^n}(x)$, so H is a homotopy from $g \circ f$ to $\mathrm{id}_{\mathbb{R}^n}$. Therefore $g \circ f \simeq \mathrm{id}_{\mathbb{R}^n}$, so f is a homotopy equivalence with homotopy inverse g. Hence $\mathbb{R}^n \simeq \{0\}$, i.e. $\mathbb{R}^n$ is contractible.

The same argument works for any $X \subset \mathbb{R}^n$ which is star-shaped about some point x_0 (e.g. any convex set), using $H(x, t) = (1 - t)x_0 + tx$.

Definition. (Deformation Retraction) Let $i : A \hookrightarrow X$ be the inclusion of a subspace. We say that A is a *deformation retract* of X if there is a map $r : X \to A$ such that $r \mid_A = \mathrm{id}_A$ and $i \circ r \simeq \mathrm{id}_X$. If the homotopy is relative to A, then A is a strong deformation retract of X.

Remark. This implies that $X \simeq A$ by the inclusion map from A to X and r from X to A. This is a strictly stronger condition.

Definition. (Retraction) A *retraction* of X to A is a map $r : X \to A$ such that $r \mid_A = \mathrm{id}$.

Remark. A retraction need not be a homotopy equivalence.

Example. X is a space and $x \in X$. Then X retracts to $\{x\}$. But $X \simeq \{x\} \iff X$ is contractible

2.2 Path

Definition. (Path) For a space X and points $x_0, x_1 \in X$, a *path* from x_0 to x_1 is a map $\gamma : I \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

We write that $\gamma : x_0 \rightsquigarrow x_1$, and if $x_0 = x_1$ we say that γ is a loop based at x_0.

If we have $\gamma : x_0 \rightsquigarrow x_1$ and $\gamma' : x_1 \rightsquigarrow x_2$ then the concatenation of $\gamma \circ \gamma'$ is defined by using the gluing lemma and going in double time which goes from x_0 to x_2.

We also define the inverse path of $\gamma : x_0 \rightsquigarrow x_1$ as $\gamma^{-1}(t) = \gamma(1-t) : x_1 \rightsquigarrow x_0$.

Finally we define the constant path $c_{x_0}(t) = x_0 : x_0 \rightsquigarrow x_0$.

If we define a relation $\sim$ on X by $x \sim y \iff$ there is a path from x to y, these definitions show that $\sim$ is an equivalence relation.

Definition. (Path components) The equivalence classes of $\sim$ are called *path components* of X. The set of all path components of X is denoted by $\pi_0(X)$. If $|\pi_0(X)| = 1$ then we say that X is *path-connected*.

Remark. X is path-connected $\iff$ $\forall x, y \in X$ there exists a path $x \rightsquigarrow y$.

Proposition. Given a map $f : X \to Y$ there is a well-defined associated function

$$\pi_0(f) : \pi_0(X) \to \pi_0(Y)$$

given by $\pi_0(f)[x] = [f(x)]$. Futhermore,

(i) If $f \simeq g$, then $\pi_0(f) = \pi_0(g)$;

(ii) For maps $A \xrightarrow{h} B \xrightarrow{k} C$ we have that

$$\pi_0(k \circ h) = \pi_0(k) \circ \pi_0(h).$$

(iii) $\pi_0(\mathrm{id}_X) = \mathrm{id}_{\pi_0(X)}$.

Proof. Choose $x, x' \in X$ such that $[x] = [x']$. Then there exists a path γ from x to x' in X. Now $f \circ \gamma$ is a path from $f(x)$ to $f(x')$ hence they are in the same path-component so $\pi_0(f)$ is well-defined.

For (i), let $H : X \times I \to Y$ be a homotopy from f to g. Choose any $x \in X$. The map $H(x, \cdot)$ is a path from $f(x)$ to $g(x)$ so $[f(x)] = [g(x)]$ and hence $\pi_0(f) = \pi_0(g)$. (ii) and (iii) are obvious. □

Corollary. If $f : X \to Y$ is a homotopy equivalence then $\pi_0(f)$ is bijective.

Proof. Let $g : Y \to X$ be the homotopy inverse of f so $fg \simeq \mathrm{id}_Y$ hence $\pi_0(f)\pi_0(g) = \pi_0(\mathrm{id}_Y) = \mathrm{id}_{\pi_0(Y)}$. Similary $gf \simeq \mathrm{id}_X$ gives the other direction. Hence $\pi_0(f)$ is a bijection. □

Example. Consider the space $\{0, 1\}$ equipped with the discrete topology. Let's prove it is not contractible. If it were, then we'd have $\pi_0(\{0,1\}) \simeq \pi_0(*)$ (the one point space). This is false because $\{0, 1\}$ has two path-components.

Example. $\mathbb{R}^2$ does not retract to $A = \{(-1,0),(1,0)\}$. Suppose that $r : \mathbb{R}^2 \to A$ is a retraction. Then the composition

$$\pi_0(A) \overset{\pi_0(\tau)}{\to} \pi_0(\mathbb{R}^2) \overset{\pi_0(r)}{\to} \pi_0(A)$$

has to be the identity. This is impossible.

Definition. (Homotopic paths) The paths $\gamma, \gamma' : I \to X$ are *homotopic as paths* if they are homotopic rel $\{0,1\} \subseteq I$.

Lemma. If $\gamma_0, \gamma_1 : x_0 \rightsquigarrow x_1$ and $\gamma_0', \gamma_1' : x_1 \rightsquigarrow x_2$ are homotopic then $\gamma_0 \circ \gamma_0' \simeq \gamma_1 \circ \gamma_1' : x_0 \rightsquigarrow x_2$.

Proof. Let H be the homotopy of paths from γ_0 to γ_1 and let H' be the homotopy of paths from γ_0' to γ_1'. Define

$$H''(s,t) = \begin{cases} H(2s,t) & 0 \le s \le \frac{1}{2} \\ H'(2s-1,t) & \frac{1}{2} \le s \le 1 \end{cases}$$

which is continuous by the gluing lemma. At $H''(s,0) = (\gamma_0 \circ \gamma_0')(s)$ and $H''(s,1) = \gamma_1 \circ \gamma_1'$. So H'' is a homotopy from $\gamma_0 \circ \gamma_0'$ to $\gamma_1 \circ \gamma_1'$.

Proposition. Let $\gamma_0 : x_0 \rightsquigarrow x_1, \gamma_1 : x_1 \rightsquigarrow x_2, \gamma_2 : x_2 \rightsquigarrow x_3$ be paths in X. Then
(i) $(\gamma_0 \circ \gamma_1) \circ \gamma_2 \simeq \gamma_0 \circ (\gamma_1 \circ \gamma_2)$;
(ii) $\gamma_0 \circ c_{x_1} \simeq \gamma_0 \simeq c_{x_0} \circ \gamma_0$;
(iii) $\gamma_0 \gamma_0^{-1} \simeq c_{x_0}$, $\gamma_0^{-1} \circ \gamma_0 \simeq c_{x_1}$.

Proof. From these, we arrive at the following formulae.

(i) The homotopy

$$H(s,t) = \begin{cases} \gamma_0\left(\dfrac{4s}{t+1}\right) & 0 \le s \le \dfrac{t+1}{4}, \\ \gamma_1(4s-1-t) & \dfrac{t+1}{4} \le s \le \dfrac{t+2}{4}, \\ \gamma_2\left(1 - \dfrac{4(1-s)}{2-t}\right) & \dfrac{t+2}{4} \le s \le 1. \end{cases}$$

goes from $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2$ to $\gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$ as paths from x_0 to x_3.

(ii) The homotopy

$$H(s,t) = \begin{cases} \gamma_0\left(\dfrac{2s}{t+1}\right) & 0 \le s \le \dfrac{t+1}{2}, \\ x_1 & \dfrac{t+1}{2} \le s \le 1, \end{cases}$$

goes from $\gamma_0 \cdot c_{x_1}$ to γ_0 as paths from x_0 to x_1.

(iii) The homotopy

$$H(s,t) = \begin{cases} \gamma_0(2s) & 0 \le s \le \dfrac{1-t}{2}, \\ \gamma_0(1-t) & \dfrac{1-t}{2} \le s \le \dfrac{1+t}{2}, \\ \gamma_0(2-2s) & \dfrac{1+t}{2} \le s \le 1. \end{cases}$$

goes from $\gamma_0 \cdot \overline{\gamma_0}$ to c_{x_0} as paths from x_0 to x_0.

Proposition. Let X be a space, $x_0 \in X$. Define

$$\pi_1(X, x_0) = \{\text{loops in } X \text{ based at } x_0\}/\text{homotopy}.$$

Then the rule $[\gamma] \circ [\gamma'] = [\gamma \circ \gamma']$ defines a group structure on $\pi_1(X, x_0)$. The identity is paths homotopic to the constant path.

Proof. The previous proposition.

Definition. (Fundamental Group) We call $\pi_1(X, x_0)$ the *fundamental group* of X at x_0.

Definition. (Based space) A *based space* is a pair (X, x_0) where x_0 is called the *basepoint*. Given based spaces $(X, x_0), (Y, y_0)$ a *based map* between them is a map $f : X \to Y$ such that $f(x_0) = y_0$. A *based homotopy* is a homotopy relative to the basepoint.

Proposition. For a based map $f : (X, x_0) \to (Y, y_0)$ there is a well-defined associated group homomorphism

$$\pi_1(f) : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

given by $\pi_1(f)[\gamma] \to [f \circ \gamma]$, with the following

(i) If f is based homotopic to g then $\pi_1(f) = \pi_1(g)$.
(ii) $\pi_1(f \circ g) = \pi_1(f) \circ \pi_1(g)$.
(iii) $\pi_1(\mathrm{id}_X) = \mathrm{id}_{\pi_1(X,x_0)}$.

Proof. $\pi_1(f)$ is well-defined because if $\gamma \simeq \gamma'$ then $f \circ \gamma \simeq f \circ \gamma'$.

We know that $f \circ c_{x_0} = c_{f(x_0)} = c_{y_0}$ so $\pi_1(f)$ preserves the identity element of the group. We also have that

$$f \circ (\gamma \cdot \gamma') = (f \circ \gamma) \cdot (f \circ \gamma')$$

so $\pi_1(f)$ is multiplicative, hence $\pi_1(f)$ is a group homomorphism.

For (i) if f, g are homotopic rel x_0 then $f \circ \gamma \simeq g \circ \gamma$ so $\pi_1(f) = \pi_1(g)$. For (ii) since $\pi_(f \circ g)[\gamma] = [f \circ g \circ \gamma] = \pi_(f)[g \circ \gamma] = \pi_1(f) \circ \pi_1(g)$. Part (iii) is obvious. □

For ease of notation, we'll just write f_* to mean $\pi_1(f)$.

Proposition. Let $u : x_0 \rightsquigarrow x_1$ be a path in X. Then it induces a group homomorphism

$$\begin{aligned} u_\# : \pi_1(X, x_0) &\to \pi_1(X, x_1) \\ [\gamma] &\to [u^{-1} \cdot \gamma \cdot u] \end{aligned}$$

satisfying

(i) $u \simeq u' \implies u_\# = u'_\#$;
(ii) $(c_{x_0})_\# = \mathrm{id}_{\pi_1(X,x_0)}$;
(iii) $v : x_1 \rightsquigarrow x_2 \implies (u \cdot v)_\# = v_\# u_\#$;
(iv) If $x_0 = x_1$, then $u_\#$ is conjugation by $[u]$;
(v) $f : X \to Y$, then we have

$$(f \circ u)_\# \circ f_* = f_* \circ u_\#.$$

Proof. Parts (i)-(iv) are obvious. We'll just prove (v).

We have that

$$\begin{aligned} ((f \circ u)_\# \circ f_*)[\gamma] &= (f \circ u)_\#[f \circ \gamma] \\ &= [(f \circ u)^{-1} \cdot (f \circ \gamma) \cdot (f \circ u)] \\ &= [f \circ (u^{-1} \cdot \gamma \cdot u)] \\ &= f_*[u^{-1} \cdot \gamma \cdot u] \\ &= f_* u_\#[\gamma]. \quad \square \end{aligned}$$

Suppose $f, g : X \to Y$ with $x_0 \in X$ and $H : X \times I \to Y$ is a homotopy from f to g. Then $H(x_0, \cdot) : I \to Y$ is a path from $f(x_0)$ to $g(x_0)$. Write this path as u. This sets us up to prove the following lemma

Lemma. For u, f, g given as above,

$$u_\# \circ f_* = g_*.$$

Proof. Choose any loop γ at x_0. Consider the composition

$$F : I \times I \xrightarrow{\gamma \times \mathrm{id}_I} X \times I \xrightarrow{H} Y$$

so $F(s,t) = H(\gamma(s), t))$. $\square$

Theorem. Let $f : X \to Y$ and choose $x_0 \in X$. Then $f_* : \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$. If f is a homotopy equivalence, then f_* is a group isomorphism.

Proof. Let $g : Y \to X$ be a homotopy inverse for f such that $f \circ g \simeq \mathrm{id}$ and $g \circ f \simeq \mathrm{id}$. Choose homotopies H from fg to id and H' from gf to id. Let

$$u'(t) = H'(x_0, 1 - t) : I \to X.$$

Then by the previous lemma with the identity map we get that $u'_\# = (g \circ f)_* = g_* \circ f_*$. We know that $u'_\#$ is an isomorphism hence $g_* \circ f_*$ and $f_* \circ g_*$ are both isomorphisms (by doing the

same thing the otherway around). Hence since the first result gives that f_* is injective and g_* is surjective, and the second result gives that the otherway around, we get that both functions are isomorphisms, namely that f_* is an isomorphism. □

Definition. (Simply connected) A space X is *simply connected* if it is path-connected and satifies $\pi_1(X, x_0) = 1$ for all $x_0 \in X$.

Remark. From the previous proposition, since X is path-connected, we can connected every two points with a path, so $\pi_1(X, x_0) \cong \pi_1(X, x_1)$, hence it's equivalent that $\pi_1(X, x_0) = 1$ for a $x_0 \in X$ is enough for equivalence of simply connectedness.

Example. Any contractible space is simply connected. This is clear since if X is contractible then $X \simeq \{*\}$ so $\pi_1(X, x_0) \cong \pi_1(*, *) = 1$.

Lemma. A space X is simply connected if and only if for all x_0, x_1 there is a unique homotopy class of paths from x_0 to x_1.

Proof. The forward direction is obvious just from taking $x_0 = x_1$ where the result follows directly.

Instead if we suppose that X is simply connected and $\pi_1(X, x_1) = 1$. Since X is path-connected, there is a path from x_0 to x_1. Choose two paths γ and γ' from x_1 to x_1. Then $[\gamma^{-1} \cdot \gamma'] \in \pi_1(X, x_1)$. So $\gamma^{-1} \cdot \gamma' \simeq c_{x_1}$. Hence there is a homotopy from $\gamma \cdot \gamma^{-1} \cdot \gamma'$ to $\gamma \cdot c_{x_1}$. Since we can reduce the path $\gamma \cdot \gamma^{-1}$ to a point, we have that $\gamma' \simeq \gamma$. □

3 Covering Space

Definition. (Covering map) A map $f : \tilde{X} \to X$ is a *covering map* if it is surjective and for all $p \in X$ there is an open neighbourhood $U_p \subset X$ such that $f^{-1}(U_p) \simeq \bigsqcup_{\alpha \in I} V_\alpha$ where each

$$f|_{V_\alpha} : V_\alpha \to U_p$$

is a homeomorphism. We say that $\tilde{X}$ is the covering space of X.

Example. Clearly homeomorphism are covering maps

Example. If $p : \tilde{X} \to X$ and $q : \tilde{Y} \to Y$ are covering maps then $p \times q : \tilde{X} \times \tilde{Y} \to X \times Y$ is a covering map.

Example. Consider $S^1 = \{z \in \mathbb{C} : |z| = 1\} \subseteq \mathbb{C}$. Take $p(t) = e^{2\pi i t} : \mathbb{R} \to S^1$.

To prove that p is a covering map, consider the set $U_{y>} = \{x + iy \in S^1 : y > 0\}$, the open upper semicircle. Then

$$p^{-1}(U_{y>0}) = \bigsqcup_{j \in \mathbb{Z}} \left(j, j + \frac{1}{2}\right) \subset \mathbb{R}$$

which is the disjoint union of $\mathbb{Z}$ many open intervals. Furthermore

$$p|_{(j, j+\frac{1}{2})} : \left(j, j + \frac{1}{2}\right) \to U_{y>0}$$

is a continuous bijection with inverse

$$U_{y>0} \longrightarrow \left(j, j+\frac{1}{2}\right)$$
$$x+iy \longrightarrow j+\cos^{-1}(x)/2\pi.$$

Similarly with $U_{y<0}, U_{x>0}, U_{x<0}$ these open sets cover S^1. Thus p is a covering map.

Example. Consider $p(z) = z^n : S^1 \to S^1$ and denote $\eta = e^{\frac{2\pi i}{n}}$ and let ζ be some nth root of y. Then we have that

$$p^{-1}(y) = \{\zeta, \eta\zeta, \eta^2\zeta, \ldots, \eta^{n-1}\zeta\}$$

and $p^{-1}(S^1 \setminus \{y\})$ is the complement of this set. Now we can define

$$V_0 = \{z \in S^1 \mid 0 < \arg(z/\eta) < 2\pi/n\}$$

and $V_i = V_0 \cdot \eta^i$ for $0 \leq i < n$, so we get that

$$p^{-1}(S^1 \setminus \{y\}) = \bigsqcup_{i=0}^{n-1} V_i.$$

Then $p\,|_{V_i} \colon V_i \to S^1 \setminus \{y\}$ is a homeomorphism and

$$S^1 = \bigcup_{y \in S^1} S^1 \setminus \{y\}$$

therefore p is a covering map.

Example. Consider $S^2 = \{v \in \mathbb{R}^3 \mid ||v|| = 1\} \subseteq \mathbb{R}^3$. We can define an equivalence relation $x \sim -x$ on S^2 which defines a quotient space $\mathbb{RP}^2 = S^2/\sim$. Consider the quotient map

$$p : S^2 \longrightarrow \mathbb{RP}^2$$
$$x \longrightarrow [x]$$

which we claim is a covering map. Let $V = \{(x, y, z|) \in S^2 \mid z \neq 0\}$ then $V \simeq V_{z>0} \sqcup V_{z<0}$ so we can define p on the restriction to each of these subspaces. We claim that

$$p\,|_{V_{z>0}} \colon V_{z>0} \to p(V)$$
$$p\,|_{V_{z<0}} \colon V_{z<0} \to p(V)$$

are homeomorphisms. Similarly define $V_x, V_y \subset S^2$ then $p(V_x), p(V_y), p(V_z)$ are evenly covered and forms san open cover of $\mathbb{RP}^2$. Hence p is a covering map.

Definition. (Lifting) Let : $p : \tilde{X} \to X$ be a covering map and $f : Y \to X$ be any map. Then a map $\tilde{f} : Y \to \tilde{X}$ is a *lift* of f along p if $f = p \circ \tilde{f}$.

Lemma. (Uniqueness of lifts lemma) Let $p : \tilde{X} \to X$ be a covering map and $f : Y \to X$ be any map. Let $\tilde{f}_0, \tilde{f}_1$ be two lifts of f along p. Then the set $\{y \in Y : \tilde{f}_0(y) = \tilde{f}_1(y)\}$ is clopen in Y. Consequently if Y is connected then $S = Y$ or $S = \emptyset$.

Proof. First we'll show that S is open. Choose $y \in S$ and let U be an open neighbourhood of $f(s) \in X$ which is evenly covered, so $p^{-1}(U) = \bigsqcup_{\alpha \in I} V_\alpha$. Then $\tilde{f}_0(y)$ and $\tilde{f}_1(y)$ are equal so lie in the same V_β. Thus we have that on $N = \tilde{f}_0^{-1}(V_\beta) \cap \tilde{f}_1^{-1}(V_\beta)$ we have that

$$p|_{V_\beta} \circ \tilde{f}_0 \mid_N = f \mid_N = p \mid_{V_\beta} \circ \tilde{f}_1 \mid_N$$

but since $p \mid V_\beta$ is a homeomorphism, it has an inverse so $\tilde{f}_0 \mid_N = \tilde{f}_1 \mid_N$ and $y \in N \subset S$, so S is open.

Now we have to show that S is closed. We'll show that $Y \setminus S$ is open. Suppose that $y \notin S$, so $\tilde{f}_0(y) \neq \tilde{f}_1(y)$. Choose an evenly covered open neighbourhood U at $f(y)$ and identify $p^{-1}(U) = \bigsqcup_{\alpha \in I} V_\alpha$ so $\tilde{f}_0(y) \in V_\alpha$ and $\tilde{f}_1(y) \in V_\beta$ for some $\alpha, \beta \in I$ since V_α is mapped homeomorphically to U and the covering maps differ, we must have that $\alpha \neq \beta$. So $N = \tilde{f}_1^{-1}(V_\beta) \cap \tilde{f}_0^{-1}(V_\alpha)$ is an open neighbourhood of y so must intersect S by the definition of closure. But then V_α and V_β must intersect which is a contradiction.

Lemma. (Homotopy lifting lemma) Let $p : \tilde{X} \to X$ be a covering map, let Y be an arbitrary topological space, and let $H : Y \times I \to X$ be a homotopy from f_0 to f_1, i.e. $H(-,0) = f_0$ and $H(-,1) = f_1$. Let $\tilde{f}_0 : Y \to \tilde{X}$ be a lift of f_0. Then there exists a unique homotopy $\tilde{H} : Y \times I \to \tilde{X}$ such that
(i) $\tilde{H}(-,0) = \tilde{f}_0$;
(ii) $\tilde{H}$ is a lift of H along p, i.e. $p \circ \tilde{H} = H$.
In particular, $\tilde{H}$ is a homotopy from $\tilde{f}_0$ to the lift $\tilde{f}_1 := \tilde{H}(-,1)$ of f_1.

Proof. Throughout, $I = [0,1]$. We use the *uniqueness of lifts* lemma in the following form: if Z is a connected space and $g_1, g_2 : Z \to \tilde{X}$ are two lifts along p of the same map $Z \to X$ which agree at one point of Z, then $g_1 = g_2$. We only ever apply it with $Z = I$; in particular no connectedness (or local path connectedness) of Y is needed anywhere below.

Step 1: uniqueness. Let $\tilde{H}_1$ and $\tilde{H}_2$ both satisfy (1) and (2). Fix $y \in Y$. The paths $t \mapsto \tilde{H}_1(y,t)$ and $t \mapsto \tilde{H}_2(y,t)$ in $\tilde{X}$ are both lifts of the path $t \mapsto H(y,t)$ in X, and by (1) both start at $\tilde{f}_0(y)$. Since I is connected, uniqueness of lifts gives $\tilde{H}_1(y,t) = \tilde{H}_2(y,t)$ for all $t \in I$. As $y \in Y$ was arbitrary, $\tilde{H}_1 = \tilde{H}_2$.

This argument works for every space Y, so it applies in particular to any open subset $Y' \subseteq Y$ with the restricted data $H|_{Y' \times I}$ and $\tilde{f}_0|_{Y'}$.

Step 2: local existence. We claim that for every $y_0 \in Y$ there exist an open neighbourhood U_{y_0} of y_0 in Y and a continuous map $\tilde{H}_{y_0} : U_{y_0} \times I \to \tilde{X}$ such that $p \circ \tilde{H}_{y_0} = H|_{U_{y_0} \times I}$ and $\tilde{H}_{y_0}(-,0) = \tilde{f}_0|_{U_{y_0}}$.

(2a) Choosing a partition. Fix $y_0 \in Y$. For each $t \in I$ the point $H(y_0,t)$ has an evenly covered open neighbourhood in X. By continuity of H there are an open neighbourhood W_t of y_0 in Y and an open neighbourhood N_t of t in I such that $H(W_t \times N_t)$ is contained in an evenly covered open subset of X. The sets $\{N_t\}_{t \in I}$ form an open cover of the compact space I, so there is a finite subcover $N_{s_1}, \ldots, N_{s_k}$. Put

$$W = W_{s_1} \cap \cdots \cap W_{s_k},$$

an open neighbourhood of y_0. By the Lebesgue number lemma there is an $N \geq 1$ such that each interval $[\frac{i}{N}, \frac{i+1}{N}]$ lies in some N_{s_j}. Write $t_i = \frac{i}{N}$. Then for each $i = 0, \ldots, N-1$ there is an evenly

covered open subset $U_i \subseteq X$ with

$$H\left(W \times [t_i, t_{i+1}]\right) \subseteq H(W_{s_j} \times N_{s_j}) \subseteq U_i.$$

Write

$$p^{-1}(U_i) = \bigsqcup_{\alpha \in A_i} V_{i,\alpha}, \qquad p\,|_{V_{i,\alpha}} \colon V_{i,\alpha} \xrightarrow{\sim} U_i$$

where the $V_{i,\alpha}$ are pairwise disjoint open subsets of $\tilde{X}$ and $\xrightarrow{\sim}$ means a homeomorphism.

(2b) Inductive construction with shrinking. We show by induction on $i = 0, \dots, N$ that there exist an open neighbourhood W_i' of y_0 with $W_i' \subseteq W$ and a continuous map $G_i : W_i' \times [0, t_i] \to \tilde{X}$ such that $p \circ G_i = H|_{W_i' \times [0,t_i]}$ and $G_i(y, 0) = \tilde{f}_0(y)$ for all $y \in W_i'$.

For $i = 0$ take $W_0' = W$ and $G_0(y, 0) = \tilde{f}_0(y)$; then $p \circ G_0 = f_0|_W = H(-, 0)|_W$.

Suppose W_i' and G_i are constructed for some $i < N$. The point $G_i(y_0, t_i)$ satisfies $p(G_i(y_0, t_i)) = H(y_0, t_i) \in U_i$, so it lies in $p^{-1}(U_i)$, hence in exactly one sheet $V = V_{i,\alpha}$. Since $y \mapsto G_i(y, t_i)$ is continuous and V is open,

$$W_{i+1}' = \{y \in W_i' \mid G_i(y, t_i) \in V\}$$

is an open neighbourhood of y_0. Define $G_{i+1} : W_{i+1}' \times [0, t_{i+1}] \to \tilde{X}$ by

$$G_{i+1}(y, t) = \begin{cases} G_i(y, t) & t \in [0, t_i], \\ (p\,|_V)^{-1}\left(H(y, t)\right) & t \in [t_i, t_{i+1}]. \end{cases}$$

The second formula makes sense because $H(y, t) \in U_i$ for $y \in W \supseteq W_{i+1}'$ and $t \in [t_i, t_{i+1}]$. The two formulas agree at $t = t_i$: indeed $G_i(y, t_i) \in V$ and $p(G_i(y, t_i)) = H(y, t_i)$, so $(p\,|_V)^{-1}(H(y, t_i)) = G_i(y, t_i)$. Both pieces are continuous on closed subsets $W_{i+1}' \times [0, t_i]$ and $W_{i+1}' \times [t_i, t_{i+1}]$ whose union is the whole domain, so G_{i+1} is continuous by the pasting lemma. Clearly $p \circ G_{i+1} = H$ and $G_{i+1}(-, 0) = \tilde{f}_0$ on W_{i+1}', which completes the induction.

Finally set $U_{y_0} = W_N'$ and $\tilde{H}_{y_0} = G_N$. Note that we shrank the neighbourhood only finitely many times, so U_{y_0} is still an open neighbourhood of y_0; this is exactly where one would otherwise want Y to be (locally) connected, and the shrinking trick avoids it.

Step 3: the local lifts are compatible. Let $y_0, y_1 \in Y$ and $Y' = U_{y_0} \cap U_{y_1}$, which is open. If $Y' = \emptyset$ there is nothing to show. Otherwise $\tilde{H}_{y_0}|_{Y' \times I}$ and $\tilde{H}_{y_1}|_{Y' \times I}$ are both lifts of $H|_{Y' \times I}$ whose restriction to $Y' \times \{0\}$ is $\tilde{f}_0|_{Y'}$. By the uniqueness in Step 1 (applied to Y'), $\tilde{H}_{y_0} = \tilde{H}_{y_1}$ on $(U_{y_0} \cap U_{y_1}) \times I$.

Step 4: gluing. Define $\tilde{H} : Y \times I \to \tilde{X}$ by

$$\tilde{H}(y, t) = \tilde{H}_{y_0}(y, t) \quad \text{for any } y_0 \in Y \text{ with } y \in U_{y_0}.$$

(Such a y_0 exists, e.g. $y_0 = y$.) This is well defined by Step 3. The sets $U_{y_0} \times I$ form an open cover of $Y \times I$ and $\tilde{H}$ restricts to the continuous map $\tilde{H}_{y_0}$ on each of them, so $\tilde{H}$ is continuous, since continuity is a local property. Moreover $p \circ \tilde{H} = H$ and $\tilde{H}(-, 0) = \tilde{f}_0$ hold on each $U_{y_0} \times I$, hence on $Y \times I$. This proves existence, and uniqueness was shown in Step 1. □

Corollary. (Path lifting) For a path $\gamma : I \to X$ choose $\tilde{x}_0 \in \tilde{X}$ such that $p(\tilde{x}_0) = \gamma(0)$. Then there exists a unique path $\tilde{\gamma} : I \to \tilde{X}$ such that $\tilde{\gamma}(0) = \tilde{x}_0$ and $p \circ \tilde{\gamma} = \gamma$.

Proof. Take $Y = \{*\}$ and apply the homotopy lifting lemma.

Corollary. Let $\gamma, \gamma' : I \to X$ be paths in X from x_0 to x_1 and $\tilde{\gamma}, \tilde{\gamma}' : I \to \tilde{X}$ be lifts starting at $\tilde{x}_0 \in p^{-1}(x_0$. If $\gamma \simeq \gamma'$ as paths, then $\tilde{\gamma} \simeq \tilde{\gamma}'$ as paths, in particular $\tilde{\gamma}(1) \simeq \tilde{\gamma}'(1)$.

Proof. Let $H : I \times I \to X$ be a homotopy of paths from γ to γ' and lifts strating at $\tilde{\gamma}$ to obtain a homtoopy $\tilde{H} : I \times I \to \tilde{X}$.